

AER404 Flight Simulator — Physics Engine

Role & Result: Physics Engine team delivered a working **6-DOF** real-time physics engine in **MATLAB/Simulink**, validated with time response plots and modularized for integration with graphics and joystick subsystems.

Project Overview

You built the core flight-dynamics engine for a simulator of a narrow-body twinjet (240–280 pax) intended to compete with the 737/A320 families. The Physics Engine group’s mandate was to model the aircraft’s six-degree-of-freedom motion for use in an integrated training environment and live visualization.

Objectives

The engine had to:

- Accept pilot inputs (dual throttle, aileron, elevator, rudder).
 - Predict Euler angles (φ , θ , ψ) and body rates (P , Q , R).
 - Compute body-axis velocities (U , V , W).
 - Propagate geodetic position (longitude, latitude, altitude).
 - Use ≥ 4 th-order Runge–Kutta integration and include aero coefficients, engine dynamics, actuator limits supplied by the aircraft design work.
 - Remain modular for integration with a graphics engine (FlightGear-style) and a joystick UI.
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Modeling & Implementation

The final model uses flat-Earth kinematics for this version (chosen for simplicity), advancing the 12-state vector with a 4th-order Runge–Kutta scheme each timestep. The core dynamics live in the MATLAB function **FLIGHTEQUATIONSNEW**, which consumes the 12 states (λ , μ , h , φ , θ , ψ , U , V , W , P , Q , R) and 5 inputs ($\delta T1$, $\delta T2$, δA , δE , δR).

Simulink architecture

A minimal, testable Simulink layout couples:

- a Constant (5×1) control-input block,
- the Aircraft subsystem (physics engine + integrators), and
- Scopes for state outputs/time histories.
- This layout made it easy to swap initial conditions, hold constant inputs, and verify state trajectories before graphics/joystick hookup.

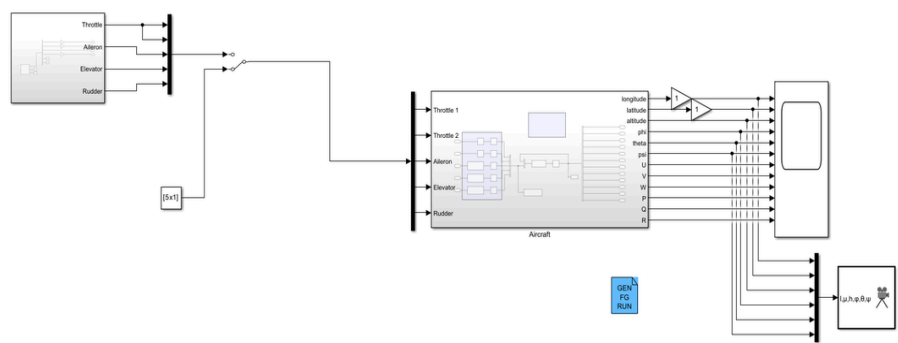


Figure 1: The Simulink simulation program's full block diagram

Test Conditions & Validation

You ran the model from realistic Toronto-area initial conditions with small attitude offsets and modest thrust:

- Lon/Lat/Alt: -79.625335° , 43.676856° , 7224 m
 - Attitude: $\varphi=0$, $\theta=0.1$ rad, $\psi=0$
 - Velocities: $U=84.9905$ m/s, $V=0$, $W=1.2713$ m/s
 - Rates: $P=0$, $Q=0.0149$ rad/s, $R=0$
 - Inputs: $\delta T1=\delta T2=0.0821$, $\delta A=0$, $\delta E=-0.1780$, $\delta R=0$
- The time-response plots for all 12 states matched expectations for the chosen inputs/ICs, validating the physics module.

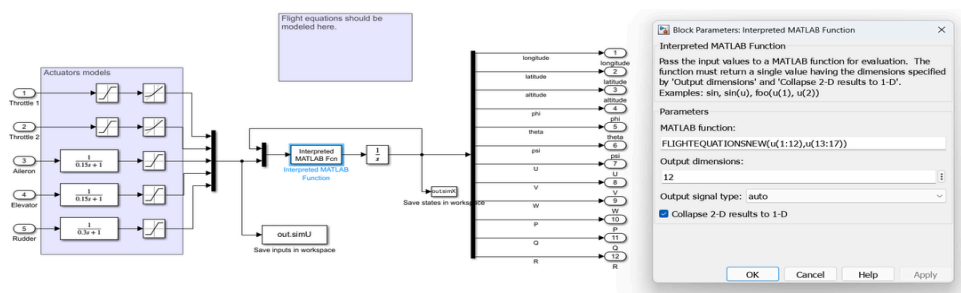


Figure 2: Aircraft subsystem block diagram\

What the Engine Captures

- Forces/Moments: aerodynamic forces (L/D/side) + engine thrust; resulting roll/pitch/yaw moments feed the body-rate equations.
 - Kinematics: body-frame derivatives mapped to NED/geodetic updates; outputs include λ , μ , $\dot{h}$, ϕ , θ , ψ , $\dot{U}$, $\dot{V}$, $\dot{W}$, $\dot{P}$, $\dot{Q}$, $\dot{R}$, which are integrated to states each step. (See [outDot](#) construction in the code.)
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Limitations & Next Steps

- Flat-Earth kinematics introduce error at long range / high altitude; upgrade to ECEF/geodetic models for higher fidelity.
 - Consider more efficient numerics to improve accuracy/performance.
 - Extend physics to include winds/turbulence/temperature, autopilot & flight-controls, and failure scenarios for a fuller training toolchain.
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Skills & Deliverables

- Skills: dynamic modeling, numerical integration (RK4), Simulink modular design, verification via time-histories, and systems integration planning.
 - Deliverables: working Simulink model, full MATLAB source ([FLIGHTEQUATIONSNEW.m](#) + run scripts), documented initial conditions, and a final report summarizing architecture and future enhancements.
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Supplementary materials

The complete technical package final report, MATLAB source (Appendix A & B: physics engine and run scripts), Simulink model, initial conditions, and references is available here: [AER404 FlightSimulator](#). This work was produced by the AER404 Physics Engine team at Toronto Metropolitan University: Gaber Soltan, Efe Ataselim, Bryant Berrio, Ryan Pulickunnel, and Kyril Couture.